

HODARI-SADIKI HUBBARD-JAMES

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Department of Physics and Astronomy, Agnes Scott College

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RESEARCH INTERESTS

Stellar Activity and Ages, Astrobiology, and Time-Domain Astronomy. My research explores three interconnected areas that advance our understanding of life in the universe. I characterize nearby K dwarf stars through spectroscopy to identify older, quieter systems that could host habitable planets. I investigate astrobiology with focus on photosynthesis under non-solar light sources, particularly how life might adapt to K and M dwarf stars. I also conduct optical and radio follow-up of unexplained astrophysical transients to constrain their origin and physical mechanisms.

ACADEMIC APPOINTMENTS

Assistant Professor of Astronomy <i>Agnes Scott College</i>	<i>July 2023- Present</i> Decatur, GA
Co-Director <i>Bradley Observatory at Agnes Scott College</i>	<i>July 2024- July 2026</i> Decatur, GA
Professor of Astronomy <i>Freedom University</i>	<i>2015- 2019, 2025- Present</i> Atlanta, GA

EDUCATION

Ph.D. in Astronomy <i>Georgia State University</i>	<i>2023</i> Atlanta, GA
Dissertation: Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs	
Advisor: Todd J. Henry	
M.S. in Physics with Astronomy Concentration <i>Georgia State University</i>	<i>2020</i> Atlanta, GA
M.S. in Biomedical Engineering <i>University of Cincinnati</i>	<i>2014</i> Cincinnati, OH
B.A. in Physics minor Biology <i>Berea College</i>	<i>2012</i> Berea, KY

GRANTS & EXTERNAL FUNDING

Funded Awards (\$448,669 awarded)

NASA COMPASS: Stellar Activity in Nearby K Dwarfs (\$375,256)	2026–2028
National Aeronautics and Space Administration	<i>Principal Investigator.</i>

NASA Space Grant – AsTropaLooza (\$27,000)	Spring 2025
Georgia Space Grant Consortium (NASA)	<i>Co-Investigator. Undergraduate-led public astronomy showcase at the Bradley Observatory.</i>

SREB Dissertation Year Award (\$20,000)	2022–2023
Southern Regional Education Board	

Provost’s Dissertation Fellowship (\$15,000) Georgia State University	2022–2023
<i>Internal Awards, Agnes Scott College</i>	
Holder Professional Development Award (\$5,000) ISLANDS Astrobiology Lab at the Bradley Observatory	2026–2027
Professional Development Committee (PDC) Award (\$2,600) Astrobiology at Agnes: Research and Conference Engagement	2025–2026
Professional Development Committee (PDC) Award (\$3,813) IAU General Assembly, Cape Town, South Africa	2024–2025
Pending & In Process	
NASA Community Project Funding – Bradley Observatory (\$683K–\$1.4M) U.S. Senate, Offices of Sen. Raphael Warnock and Sen. Jon Ossoff	FY2027 <i>Lead author. Advanced to Senate Appropriations by both Georgia senators; pending.</i>

AWARDS & HONORS	
Kavli Fellow, Frontiers of Science Symposium National Academy of Sciences	2026
IAU-OAD Recommended List: <i>Stars Without Borders</i> International Astronomical Union, Office of Astronomy for Development	2026
Beth Brown Memorial Award (Honorable Mention) National Society of Black Physicists (NSBP)	Nov. 2022
Chambliss Astronomy Achievement Award at AAS 240 American Astronomical Society	Jun. 2022
Most Inspirational Alumnus Award Berea College African Students Association	2016
Waldemar Noll Prize in Physics Berea College	2012

PUBLICATIONS * *	 ORCID:0000-0003-4568-2079
First-Author & Lead Publications	
1. The Solar Neighborhood LV:Spectral Characterization of an Equatorial Sample of 580 K Dwarfs Hubbard-James, H.-S., ... Arbogast, A. (Student Mentee), et.al., 2026, AJ 171 175. 2. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs Hubbard-James, H.-S., 2024, Proceedings of the International Astronomical Union, 20(S393), 144-149. 3. The Solar Neighborhood L: Spectroscopic Discovery of K Dwarfs Younger than 1 Gyr and New Binaries within 30 Parsecs Hubbard-James, H.-S., Lesley, D.X., Henry, T.J., et al., 2022, AJ 164 174.	
Contributing Author Publications	

4. Visual Orbits of Spectroscopic Binaries with the CHARA Array. V. HD 210763 and HD 221950
Lester K. V., Singleton J., ... **Hubbard-James, H.-S.**, et al., 2026, AJ, in press.
5. Arecibo Wow! II: Revised Properties of the Wow! Signal from Archival Ohio SETI Data
Méndez A., Ortiz Ceballos K. N., Zuluaga J. I., ... **Hubbard-James, H.-S.**, **Le, M. (Student Mentee)**, et al., 2026, ApJ, Submitted.
6. A Radio Loud Quiescent K Dwarf
Frail, D., Hyman, S., Silverstein, M., ... **Hubbard-James, H.-S.**, **Byam, J. (Student Mentee)**, et al., 2025, ApJ, 989, 186.
7. Mind the Gap. I. H α Activity of M Dwarfs Near the Partially/Fully Convective Boundary and a New H α Emission Deficiency Zone on the Main Sequence
Jao, W.-C., Henry, T. J., White, R. J., ... **Hubbard-James, H.-S.**, et al., 2023, AJ, 166, 63.
8. Visual Orbits of Spectroscopic Binaries with the CHARA Array. IV. HD 61859, HD 89822, HD 109510, and HD 191692
Lester K. V., Schaefer G. H., ... **Hubbard-James, H.-S.**, et al., 2022, AJ, 164, 228.
9. The Solar Neighborhood XLVIII: Nine Giant Planets Orbiting Nearby K Dwarfs, and the CHIRON Spectrograph's Radial Velocity Performance
Paredes L. A., Henry T. J., Quinn S. N., Gies D. R., Hinojosa-Góñi R., **James H.-S.**, Jao W.-C., et al., 2021, AJ, 162, 176.
10. Mapping out the Stellar Populations of IC 2602 and IC 2391
Nisak, A.H., White, R.J., Yep, A., ... **James, H.-S.**, et al., 2022, AJ, 163, 278.
11. TOI 564 b and TOI 905 b: Grazing and Fully Transiting Hot Jupiters Discovered by TESS
Davis A. B., Wang S., Jones M., Eastman J. D., ... **James, H.-S.**, et al., 2020, AJ, 160, 229.
12. TOI 694b and TIC 220568520b: Two Low-mass Companions near the Hydrogen-burning Mass Limit Orbiting Sun-like Stars
Mireles I., Shporer A., Grieves N., Zhou G. ... **James, H.-S.**, et al., 2020, AJ, 160, 133.
13. KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS
Rodríguez Martínez R., Gaudi B. S., ... **James, H.-S.**, et al., 2020, AJ, 160, 111.
14. A Well-aligned Orbit for the 45 Myr-old Transiting Neptune DS Tuc Ab
Zhou G., Winn J. N., Newton E. R., Quinn S. N., ... **James, H.-S.**, et al., 2020, ApJL, 892, L21.
15. TESS Spots a Hot Jupiter with an Inner Transiting Neptune
Huang C. X., Quinn S. N., Vanderburg A., ... **James, H.-S.**, et al., 2020, ApJL, 892, L7.
16. MASCARA-4 b/bRing-1 b: A retrograde hot Jupiter around a bright A-type star
Dorval P., Talens G. J. J., Otten G. P. P. L., ... **James, H.-S.**, et al., 2020, A&A, 635, A60.
17. TESS discovery of an ultra-short-period planet around the nearby M dwarf LHS 3844
Vanderspek R., Huang C. X., Vanderburg A., Ricker G. R., Latham D. W., Seager S., Winn J. N., ... **James, H.-S.**, et al., 2019, ApJL, 871, L24.
18. HD2685 b: A Hot-Jupiter orbiting an early F-type star detected by TESS
Jones M. I., Brahm R., Espinoza N., Wang S., ... **James, H.-S.**, et al., 2019, A&A, 625, A16.
19. HD 202772A b: A Transiting Hot Jupiter around a Bright, Mildly Evolved Star in a Visual Binary Discovered by TESS
Wang S., Jones M., Shporer A., Fulton B. J., ... **James, H.-S.**, et al., 2019, AJ, 157, 51.

20. Rare earth doped downshifting glass ceramics for photovoltaic applications
Leonard, R.L., Gray, S.K., **James, H.-S.**, et al., 2013, Journal of non-crystalline solids, 366, pp.1-5.

TEACHING EXPERIENCE

Instructor of Record

Agnes Scott College

Fall 2023- Present

Decatur, GA

Astronomy Courses:

- AST 120: Introduction to Astronomy (The Solar System) (*Fall 2026*)
- AST 200L: Intermediate Observational Techniques (*Spring 2024, 2025, 2026*)
- AST 300: Astrophysics I: Radiation (*Fall 2023*)
- AST 301: Astrophysics II: Dynamics (*Fall 2024, 2026*)
- GBL 102/103: Global Journeys: Astronomy and Local Communities– Puerto Rico (*Spring 2026, 2027*)

Physics & Mathematics Courses:

- PHY/MAT 131: Intro to Computer Programming (*Fall 2023 & Spring 2025*)
- PHY/MAT 231: Think Like a Data Scientist (*Spring 2024, 2026, 2027*)
- PHY 205: Math for Physicists & Engineers (*Spring 2025*)
- PHY 400: Physics Capstone Colloquium (*Fall 2024, 2026*)
- PHY 420: Advanced Seminar in Physics & Astronomy (*Fall 2024, 2026*)
- PHY 503L: Elements of Physics II Lab (*Spring 2024, 2025, 2026*)

Instructor of Record

Freedom University

2018- 2019, Fall 2025

Atlanta, GA

- STEM: Astronomy (*Fall 2025*) – virtual introductory astronomy course
- SAT Mathematics Preparation (*2018 & 2019*)

Joint Instructor

Georgia State University

Spring 2022

Atlanta, GA

- Co-Lecturer for a Synchronous (Online and In-Person Instruction) Introductory Astronomy Course
ASTR 1000 (Introduction to the Universe)

Graduate Teaching Assistant

Georgia State University

Aug 2017 - Jul 2021

Atlanta, GA

- Instructor for Introductory Astronomy Lab ASTR 1010 (Astronomy of the Solar System)
Fall 2017, 2018, 2019, & 2020, Spring 2019, & 2021
- Instructor for Introductory Astronomy Lab ASTR 1020 (Stellar and Galactic Astronomy)
Fall 2020, Spring 2018, & 2020

SERVICE & PROFESSIONAL ACTIVITIES

Professional Service

Conference Programming Committee, NSBP/NSHP Joint National Meeting 2026–2027
National Society of Black Physicists / National Society of Hispanic Physicists *Co-organizing ASTRO sessions for the joint national meeting, Raleigh, NC.*

SARA Institutional Representative 2024–Present
Southeastern Association for Research in Astronomy (SARA)

Co-Organizer, AAS Division on Dynamical Astronomy Meeting 2024–2025
Georgia Institute of Technology *Assisted in planning and organizing national scientific meeting held at Georgia Tech.*

Judge, NSBP/NSHP Joint National Meeting Nov. 2024
National Society of Black Physicists / National Society of Hispanic Physicists *Served as judge for undergraduate and graduate research presentations.*

Astronomy Curriculum Committee Member Sept 2021 - June 2023
Georgia State University

CHIRON Data Manager Dec 2018 - June 2023
Georgia State University

Institutional Service

NCAA III - Faculty Athletics Representative (FAR) Sept. 2024- Present
Agnes Scott College

Faculty Advisor, Agnes Scott College SPS Aug 2023 - Aug 2025
Agnes Scott College

Research Advisor, STEM Scholars Program Summer 2024 & 2025
Agnes Scott College

Physics Faculty Search Committee Aug 2023 - May 2024
Agnes Scott College

Outreach & Public Engagement

Co-Organizer, AsTropaLooza: The Astronomy of ATL Showcase Spring 2025
Bradley Observatory, Agnes Scott College *Public outreach event funded by NASA Space Grant with student-led programming.*

Organizer & Participant, Planetary Reach Workshop Fall 2024
Bradley Observatory, Agnes Scott College *Workshop focused on developing equitable STEM outreach strategies for Black and Brown youth.*

STEM Hands Summer Camp Jun 2022
Georgia Outreach Team for Space (GOT Space)

NASA Science Activation Planetary Reach Workshop Apr 2022
Arizona State University

STUDENT MENTORING & SUPERVISION

Current Undergraduate Mentees:

Makaila Jennings - Undergraduate Student, Agnes Scott College. <i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i> <i>Placement: NSBP Simmons Research Program, Summer 2026</i>	Jan 2026 - Present
Lia Faulks - Undergraduate Student, Agnes Scott College. <i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i>	Jan 2026 - Present
Kayla Gossett Roper - Undergraduate Student, Agnes Scott College. <i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i>	May 2025 - Present
Sasha Arbogast - Undergraduate Student, Agnes Scott College. <i>Research focus: Radial velocity confirmation of K dwarf companions and exoplanets</i> <i>Co-author on AJ publication; Cornell REU, Roman Coronagraph scheduling, Summer 2026</i>	May 2025 - Present
Mai Le - Post-baccalaureate Researcher, Agnes Scott College. <i>Research focus: Optical archival analysis of the Wow! Signal field</i> <i>Co-author on ApJ submission</i>	Aug 2024 - Present

Current Graduate Student Mentees:

Sebastian Carrazco Gaxiola - PhD Candidate, Georgia State University. <i>Research focus: Stellar spectroscopy and K dwarf characterization</i> <i>Co-author on AJ publication</i>	Aug 2022 - Present
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Former Mentees:

Jacinda Byam - Undergraduate Student, College of Charleston <i>Co-author on ApJ publication</i>	May 2024 - Aug 2025
Nia Suitt - Graduate Student, University of Central Florida.	Jan 2024 - May 2025
Daniela Garcia-Lara - Undergraduate Student, Agnes Scott College.	May 2024 - May 2025
Tim Johns - Graduate Student, Georgia State University.	May 2022 - July 2023
D. Xavier Lesley - Graduate Student, Ohio State University. <i>Co-author on AJ publication</i>	Jan 2021 - July 2023
Dan Johns - Graduate Student, Georgia State University.	Sep 2020 - May 2022
Andrey Brevett - Undergraduate Student, Berea College.	Aug 2018 - Aug 2022
Edgar Ortiz - High School Student, Freedom University.	Aug 2015 - Aug 2016
Chukwuemeka Chikelu - Graduate Student, University of Cincinnati.	Aug 2014 - Aug 2015

PROFESSIONAL AFFILIATIONS

Southern Regional Education Board (SREB)	Aug 2022 - Present
National Society of Black Physicists (NSBP)	Aug 2021 - Present
American Astronomical Society (AAS)	Oct 2017 - Present
International Astronomical Union (IAU)	Aug 2023 - Present
Biomedical Engineering Society (BMES)	May 2015 - May 2017

TECHNICAL PROFICIENCY

Programming Languages

- Expert: Python, Matlab, bash
- Intermediate Proficiency: IDL, R, HTML/CSS
- Limited Proficiency: C++, Julia

Software and Packages

- Astronomy: Astropy, IRAF/PyRAF, TOPCAT, DS9
- Telescope Operation: CTIO 1.5 telescope (computerized), CTIO 0.9 telescope (computerized), Multiple Tripod & Table Top Telescopes (manual)
- Others: LaTeX, MS Office, Git Repository
- AI & Agentic Tools: Claude Code, OpenAI Codex, and LLM APIs; build and deploy agent-based workflows for research pipelines, data analysis, and course material development

INVITED TALKS & PRESENTATIONS

Commencement Address Freedom University	May 2026
Invited Talk: Finding the K Dwarf Advantage, RECONSaganza Georgia State University	Aug. 2026
Invited Talk: Dept Physics & Astronomy Colloquium Georgia State University	July. 2025
Invited Talk: Planetary Science & Astrobiology Seminar Georgia Institute of Technology	Oct. 2024
Invited Talk: Dept Physics & Astronomy Colloquium Georgia College & State University	June 2024
Invited Talk at the Bradley Observatory Open House Agnes Scott College	Dec. 2022

SELECTED CONFERENCE PRESENTATIONS

1. The K Dwarf Advantage: Stellar Activity, Ages, and the Search for Habitable Worlds
Hubbard-James, H.-S., 2026, Poster presentation, Kavli Frontiers of Science Symposium, National Academy of Sciences, Irvine, CA.
2. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs
Hubbard-James, H., Gaxiola, S., Henry, T., Lesley, D.X., Paredes, L. A., Nisak, A., 2024, Poster presentation, 32nd General Assembly International Union (IAUGA 2024).
3. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs
Hubbard-James, H., Gaxiola, S., Henry, T., Lesley, D.X., Paredes, L. A., Nisak, A., 2024, Oral presentation, 2024 Astrobiology Science Conference (AbSciCon 2024).
4. Spectral Characterization of a Complete Equatorial Sample of 665 K Dwarfs within 33 Parsecs — Active Stars, Calm Stars, and the Best Places for Habitable Worlds
Hubbard-James, H., 2023, Oral presentation, American Astronomical Society 242nd Meeting.

5. Spectroscopic Identification of Young and Active K Dwarfs Within 25 Parsecs
Hubbard-James, H., Lesley, D.X., Henry, T.J., Paredes, L. A., Nisak, A., 2022, Poster presentation, American Astronomical Society 240th Meeting Abstracts.
6. Spectroscopic Identification of Young and Active K Dwarfs Within 25 Parsecs
Hubbard-James, H., Lesley, D.X., Henry, T.J., Paredes, L. A., Nisak, A., 2022, Oral presentation, 2022 Astrobiology Science Conference (AbSciCon 2022).
7. Spectroscopic Identification of Five K Dwarfs Younger than 1 GYR Within 30 parsecs
Hubbard-James, H., Lesley, D.X., Henry, T.J., 2021, Oral presentation, 2021 National Society of Black Physicists Conference.

OTHER PROFESSIONAL EXPERIENCE

GRE and SAT Instructor, University of Georgia/Educational Testing Consultants	2019-2020
Lab Technologist, Labsolutions	2017
Biomedical Engineer, Castle Medical	2015-2016
Graduate Research Assistant, University of Cincinnati	2012-2014

PROFESSIONAL REFERENCES

Todd J. Henry, PhD	<i>Doctoral Advisor</i>
Distinguished Professor of Astronomy, Georgia State University	thenry88@gsu.edu
Director, REsearch Consortium On Nearby Stars (RECONS) and SMARTS	
Abel Méndez, PhD	<i>Research Collaborator</i>
Professor, University of Puerto Rico at Arecibo	abel.mendez@upr.edu
Director, Planetary Habitability Laboratory	